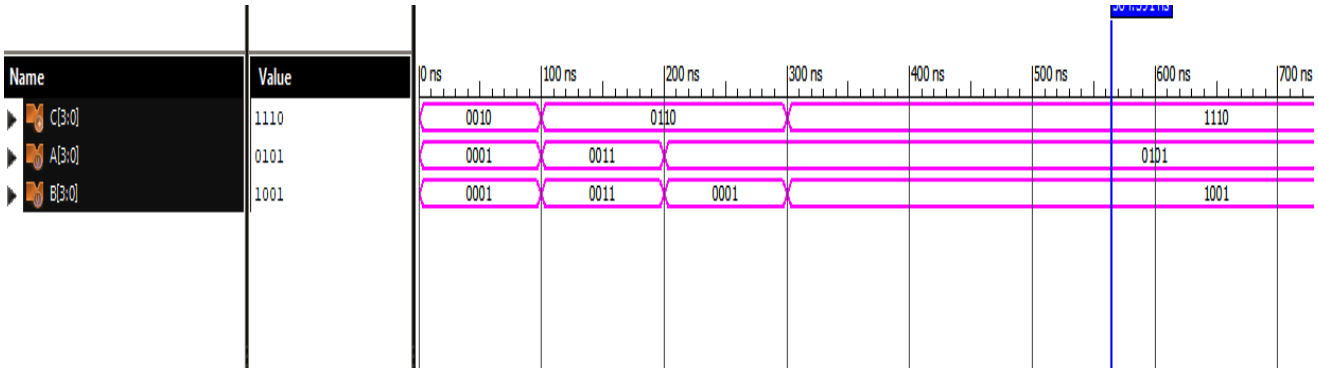
4 BIT BINARY ADDER

```
module BINARYADD(A,B,C);
input[3:0]A,B;
output[3:0]C;
assign C=A+B;
endmodule
```

4 BIT BINARY ADDER OUTPUT



VERILOG TEST FIXTURE

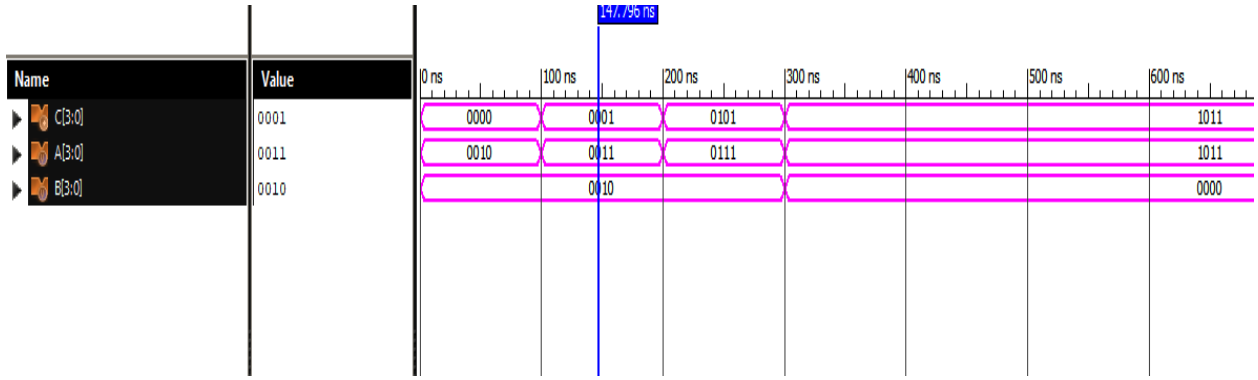
```
BINARYADD uut (
        .A(A),
        .B(B),
        .C(C)
    );

    initial begin
        A = 4'b0001;
        B = 4'b0001;
        #100;
            A = 4'b0011;
        B = 4'b0011;
        #100;
        A = 4'b0101;
        B = 4'b0001;
        #100;
        A = 4'b0101;
        B = 4'b1001;
            #100;
    End
```

---

4 BIT BINARY SUBTRACTOR

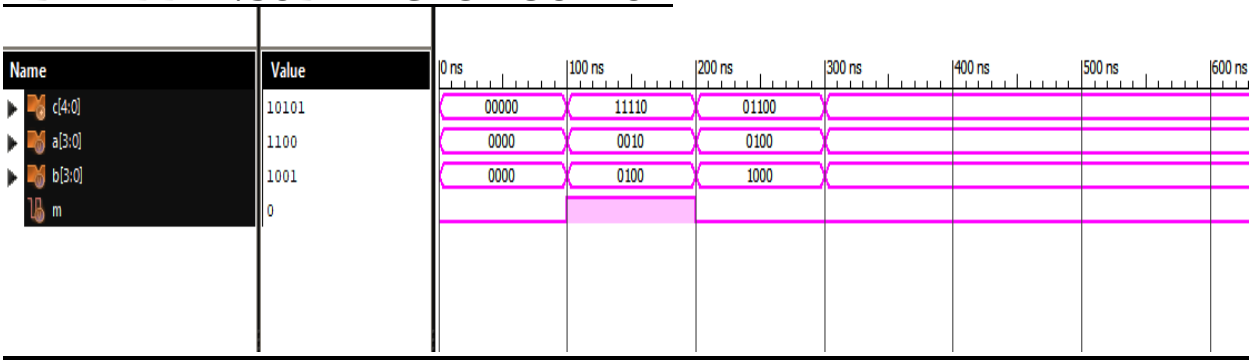
```
module BINARYADD(A,B,C);
input[3:0]A,B;
output[3:0]C;
assign C=A-B;
endmodule
```

OUTPUT 4 BIT BINARY SUBTRACTOR



---

4 BIT ADDER/SUBTRACTOR

```
module Add_Sub(a, b, m, c);
input [3:0] a, b;
input m;
output reg [4:0] c;
always @(a, b, m)
begin
   if(m == 0)
      c = a + b;
   else
      c = a - b;
end
endmodule
```

4 BIT ADDER/SUBTRACTOR OUTPUT